

Network Administration / System Administration

Homework 2

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0. VM Setup

Reference:

- [unable to retrieve community.db: 404 errors \(Arch Linux Forums\)](#)
- [sudo pacman -Syu fails with error: failed to commit transaction \(Stack Exchange\)](#)
- [OpenSSH \(ArchWiki\)](#)
- [Enable SSH on Arch Linux](#)

為了方便連線，我打算在 VM 當中配置 ssh server，使我可以利用 ssh 的方式進入虛擬機。步驟如下：

1. 執行 `run_vm.sh`，以 vnc 登入 VM 當中的 root 帳號。
2. 更新 archlinux 失敗，經查閱資料後，發現需要將 `/etc/pacman.conf` 中的 `[[community]]` 註解掉。
3. 更新 archlinux keyring

```
1 pacman -Sy archlinux-keyring
```

4. 更新 archlinux 並安裝 openssh：

```
1 pacman -Syu
2 pacman -S openssh
```

5. 啟動 sshd

```
1 systemctl enable sshd
2 systemctl start sshd
```

6. 離開 VM，並在 host 端修改 `run_vm.sh`，加入最後一行，使 host 端的 4321 port 轉發到 VM 的 22 port。

```
1  #!/bin/bash
2
3  echo "Starting VM at port $1"
4  echo "parameter is $((($1 - 5900))"
5
6  qemu-system-x86_64 -m 2G --enable-kvm \
7      -vnc :"$((($1 - 5900))",password=on -monitor stdio \
8      -drive file=disk1.qcow2,media=disk,index=0 \
9      -drive file=disk2.qcow2,if=virtio,media=disk,index=1 \
10     -drive file=disk3.qcow2,if=virtio,media=disk,index=2 \
11     -drive file=disk4.qcow2,if=virtio,media=disk,index=3 \
12     -drive file=disk5.qcow2,if=virtio,media=disk,index=4 \
13     -drive file=disk6.qcow2,if=virtio,media=disk,index=5 \
14     -drive file=disk7.qcow2,if=virtio,media=disk,index=6 \
15     -drive file=disk8.qcow2,if=virtio,media=disk,index=7 \
16     -drive file=disk9.qcow2,if=virtio,media=disk,index=8 \
17     -drive file=disk10.qcow2,if=virtio,media=disk,index=9 \
18     -net user,hostfwd=tcp::4321-:22 -net nic
```

7. 成功從工作站當中以 balu 的帳號 ssh 進去 VM。

```
1  ssh balu@localhost -p 4321
```

1. 那傢伙竟然敢無視窗

Reference:

- [NTFS3](#)
- [Journaling file system \(Wikipedia\)](#)
- [How to mount NTFS in Linux without fuseblk \(Stack Overflow\)](#)
- [使用 ntfs3 驱动替换 ntfs-3g 挂载 windows NTFS 分区](#)
- [对 NTFS 硬盘进行读写操作；开机自动挂载硬盘 \(CSDN\)](#)

NTFS 檔案系統可以滿足題目所要求，在雙系統中支援原生讀寫，並且在 Windows 11 下支援日誌功能。為了格式化 NTFS 檔案系統，需要先安裝 `ntfs-3g` 套件。我完成此題的步驟如下：

1. 在 VM 當中安裝 `ntfs-3g` 套件

```
1 sudo pacman -S ntfs-3g
```

2. 創立掛載點

```
1 sudo mkdir /mnt/myusb
```

3. 格式化硬碟

```
1 sudo mkfs.ntfs /dev/vdi2
```

4. 取得 UUID

```
1 sudo blkid /dev/vdi2
```

```
[balu@archlinux ~]$ sudo blkid /dev/vdi2
/dev/vdi2: BLOCK_SIZE="512" UUID="4E800CA236A749BE" TYPE="ntfs" PARTUUID="eb45fe31-f8c2-fd42-af97-b614bbf57e9b"
[balu@archlinux ~]$
```

Figure 1: Screenshot of `sudo blkid /dev/vdi2`

5. 配置自動掛載硬碟: 修改 `/etc/fstab`

6. 重開機，即可看到硬碟已經掛載

Ans.

```
[balu@archlinux ~]$ cat /etc/fstab
# Static information about the filesystems.
# See fstab(5) for details.

# <file system> <dir> <type> <options> <dump> <pass>
# /dev/sda2
UUID=d1daff5a-54da-43b8-a88e-83fa4e94a0b1 / ext4 rw,relatime 0 1

# /dev/sda1
UUID=711c-6167 /boot vfat rw,relatime,fmask=0022,dmask=0022,codepage=437,ioccharset=ascii,shortname=mixed,utf8,errors=remount-ro 0 2

# /dev/vdi2
UUID=4E800CA236A749BE /mnt/myusb ntfs3 defaults 0 0

/dev/nasahw2-main/course /home/balu/course ext4 defaults 0 2
/dev/nasahw2-secondary/videos /home/balu/videos ext4 defaults 0 2
[balu@archlinux ~]$
```

Figure 2: Screenshot of `cat /etc/fstab`

```

[balu@archlinux ~]$ lsblk; df -hT
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
fd0                                2:0      1   4K  0 disk
sda                                8:0      0  32G  0 disk
├─sda1                             8:1      0 200M  0 part /boot
├─sda2                             8:2      0 31.8G  0 part /
sr0                               11:0     1 1024M  0 rom
zram0                             253:0    0   983M  0 disk [SWAP]
vda                               254:0    0    1G  0 disk
├─vda1                             254:1    0 1022M  0 part
├─┌nasahw2--main-course            252:0    0  500M  0 lvm  /home/balu/course
└─vdb                             254:16   0    1G  0 disk
├─vdb1                             254:17   0 1022M  0 part
├─vdc                             254:32   0    2G  0 disk
├─vdc1                             254:33   0    2G  0 part
├─vdd                             254:48   0   16G  0 disk
├─vdd1                             254:49   0   16G  0 part
├─vde                             254:64   0   512M  0 disk
├─vde1                             254:65   0   510M  0 part
├─┌nasahw2--secondary-videos      252:1    0  508M  0 lvm  /home/balu/videos
└─vdf                             254:80   0    7G  0 disk
├─vdf1                             254:81   0    2G  0 part
├─vdf2                             254:82   0    2G  0 part
├─vdf3                             254:83   0    2G  0 part
├─vdg                             254:96   0    7G  0 disk
├─vdg1                             254:97   0    2G  0 part
├─vdg2                             254:98   0    2G  0 part
├─vdg3                             254:99   0    2G  0 part
├─vdh                             254:112  0    7G  0 disk
├─vdh1                             254:113  0    2G  0 part
├─vdh2                             254:114  0    2G  0 part
├─vdh3                             254:115  0    2G  0 part
├─vdi                             254:128  0    6G  0 disk
├─vdi1                             254:129  0    2G  0 part
└─vdi2                             254:130  0    4G  0 part /mnt/myusb
Filesystem                        Type      Size  Used Avail Use% Mounted on
dev                              devtmpfs  976M    0  976M   0% /dev
run                              tmpfs     984M  708K  983M   1% /run
/dev/sda2                       ext4      32G   2.6G   27G   9% /
tmpfs                           tmpfs     984M    0  984M   0% /dev/shm
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                           tmpfs     984M    0  984M   0% /tmp
/dev/vdi2                       ntfs3     4.0G   22M   4.0G   1% /mnt/myusb
/dev/sda1                       vfat      197M   69M  129M  35% /boot
/dev/mapper/nasahw2--main-course ext4      459M   4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4     466M   66M  371M  16% /home/balu/videos
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs                           tmpfs     197M   4.0K  197M   1% /run/user/1000
[balu@archlinux ~]$

```

Figure 3: Screenshot of `lsblk; df -hT`

2. 因為要換到新的 SWAP

Reference:

- [Create a Linux Swap File \(Linuxize\)](#)

新增 SWAP 檔案的步驟如下：

1. 建立 /newswap 檔案

```
1 sudo fallocate -l 4G /newswap
```

2. 設定檔案權限

```
1 sudo chmod 600 /newswap
```

3. 格式化檔案

```
1 sudo mkswap /newswap
```

4. 啟用 SWAP 檔案

```
1 sudo swapon /newswap
```

Ans.



```
[balu@archlinux ~]$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	207Mi	1.7Gi	4.6Mi	103Mi	1.7Gi
Swap:	5.0Gi	0B	5.0Gi			

```
[balu@archlinux ~]$
```

Figure 4: Screenshot of `free -h`



3. 為資料創造新的棲身之處

Reference:

- [lvextend 命令 \(LinuxCool\)](#)

使用指令如下：

```
1 sudo lvextend -L 1GiB /dev/nasahw2-main/course
```

Ans.

```
[root@archlinux ~]# lsblk; df -hT
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
fd0	2:0	1	4K	0	disk	
sda	8:0	0	32G	0	disk	
└─sda1	8:1	0	200M	0	part	/boot
└─sda2	8:2	0	31.8G	0	part	/
sr0	11:0	1	1024M	0	rom	
zram0	253:0	0	983M	0	disk	[SWAP]
vda	254:0	0	1G	0	disk	
└─vda1	254:1	0	1022M	0	part	
└─┬─nasahw2--main-course	252:0	0	1G	0	lvm	/home/balu/course
vdb	254:16	0	1G	0	disk	
└─vdb1	254:17	0	1022M	0	part	
└─┬─nasahw2--main-course	252:0	0	1G	0	lvm	/home/balu/course
vdc	254:32	0	2G	0	disk	
└─vdc1	254:33	0	2G	0	part	
vdd	254:48	0	16G	0	disk	
└─vdd1	254:49	0	16G	0	part	
vde	254:64	0	512M	0	disk	
└─vde1	254:65	0	510M	0	part	
└─┬─nasahw2--secondary-videos	252:1	0	508M	0	lvm	/home/balu/videos
vdf	254:80	0	7G	0	disk	
└─vdf1	254:81	0	2G	0	part	
└─vdf2	254:82	0	2G	0	part	
└─vdf3	254:83	0	2G	0	part	
vdg	254:96	0	7G	0	disk	
└─vdg1	254:97	0	2G	0	part	
└─vdg2	254:98	0	2G	0	part	
└─vdg3	254:99	0	2G	0	part	
vdh	254:112	0	7G	0	disk	
└─vdh1	254:113	0	2G	0	part	
└─vdh2	254:114	0	2G	0	part	
└─vdh3	254:115	0	2G	0	part	
vdi	254:128	0	6G	0	disk	
└─vdi1	254:129	0	2G	0	part	
└─vdi2	254:130	0	4G	0	part	/mnt/myusb

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
dev	devtmpfs	976M	0	976M	0%	/dev
run	tmpfs	984M	712K	983M	1%	/run
/dev/sda2	ext4	32G	6.6G	23G	23%	/
tmpfs	tmpfs	984M	0	984M	0%	/dev/shm
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
tmpfs	tmpfs	984M	0	984M	0%	/tmp
/dev/vdi2	ntfs3	4.0G	22M	4.0G	1%	/mnt/myusb
/dev/sda1	vfat	197M	69M	129M	35%	/boot
/dev/mapper/nasahw2--main-course	ext4	459M	4.5M	425M	2%	/home/balu/course
/dev/mapper/nasahw2--secondary-videos	ext4	466M	66M	371M	16%	/home/balu/videos
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	tmpfs	197M	4.0K	197M	1%	/run/user/1000

```
[root@archlinux ~]#
```

Figure 5: Screenshot of `lsblk; df -hT`

4. 我有拜託妳別把我的作業告訴其他人了吧

Reference:

- [How to Encrypt a Drive in Linux \(Medium\)](#)
- [How to add a passphrase, key, or keyfile to an existing LUKS device \(Red Hat\)](#)
- [Linux 使用 LUKS 加密 Ext4 檔案系統](#)

我做的步驟如下：

1. 以 root 進入 VM。

2. 切出一塊新的 lvm，作為加密區域

```
1 lvcreate -L 800MiB -n homework nasahw2-main
```

3. 加密 /dev/nasahw2-main/homework

```
1 cryptsetup luksFormat /dev/nasahw2-main/homework
```

4. 開啟加密區域

```
1 cryptsetup open /dev/nasahw2-main/homework homework
```

5. 格式化加密區域

```
1 mkfs.ext4 /dev/mapper/homework
```

6. 掛載加密區域

```
1 mount /dev/mapper/homework /home/balu/homework/
```

7. 設定 keyfile

```
1 cryptsetup luksAddKey /dev/nasahw2-main/homework /home/balu/lvm_key
```

8. 設定開機自動解密，在 /etc/crypttab 中加入底下這行：

```
1 homework /dev/nasahw2-main/homework /home/balu/lvm_key
```

9. 設定開機自動 mount，在 /etc/fstab 中加入底下這行：

```
1 /dev/mapper/homework /home/balu/homework ext4 defaults 0 2
```

10. 重開機，確認加密區域已經掛載

Ans.

```
[root@archlinux balu]# lsblk; df -hT
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
fd0	2:0	1	4K	0	disk	
sda	8:0	0	32G	0	disk	
└sda1	8:1	0	200M	0	part	/boot
└sda2	8:2	0	31.8G	0	part	/
sr0	11:0	1	1024M	0	rom	
zram0	253:0	0	983M	0	disk	[SWAP]
vda	254:0	0	1G	0	disk	
└vda1	254:1	0	1022M	0	part	
└└nasahw2--main-course	252:0	0	1G	0	lvm	/home/balu/course
vdb	254:16	0	1G	0	disk	
└vdb1	254:17	0	1022M	0	part	
└└nasahw2--main-course	252:0	0	1G	0	lvm	/home/balu/course
└└└nasahw2--main-homework	252:1	0	800M	0	lvm	
└└└└homework	252:3	0	784M	0	crypt	/home/balu/homework
vdc	254:32	0	2G	0	disk	
└vdc1	254:33	0	2G	0	part	
vdd	254:48	0	16G	0	disk	
└vdd1	254:49	0	16G	0	part	
vde	254:64	0	512M	0	disk	
└vde1	254:65	0	510M	0	part	
└└nasahw2--secondary-videos	252:2	0	508M	0	lvm	/home/balu/videos
vdf	254:80	0	7G	0	disk	
└vdf1	254:81	0	2G	0	part	
└vdf2	254:82	0	2G	0	part	
└vdf3	254:83	0	2G	0	part	
vdg	254:96	0	7G	0	disk	
└vdg1	254:97	0	2G	0	part	
└vdg2	254:98	0	2G	0	part	
└vdg3	254:99	0	2G	0	part	
vdh	254:112	0	7G	0	disk	
└vdh1	254:113	0	2G	0	part	
└vdh2	254:114	0	2G	0	part	
└vdh3	254:115	0	2G	0	part	
vdi	254:128	0	6G	0	disk	
└vdi1	254:129	0	2G	0	part	
└vdi2	254:130	0	4G	0	part	/mnt/myusb

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
dev	devtmpfs	976M	0	976M	0%	/dev
run	tmpfs	984M	732K	983M	1%	/run
/dev/sda2	ext4	32G	6.6G	23G	23%	/
tmpfs	tmpfs	984M	0	984M	0%	/dev/shm
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-networkd.service
tmpfs	tmpfs	984M	0	984M	0%	/tmp
/dev/vdi2	ntfs3	4.0G	22M	4.0G	1%	/mnt/myusb
/dev/sda1	vfat	197M	69M	129M	35%	/boot
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/systemd-cryptsetup@homework.service
/dev/mapper/nasahw2--main-course	ext4	459M	4.5M	425M	2%	/home/balu/course
/dev/mapper/nasahw2--secondary-videos	ext4	466M	66M	371M	16%	/home/balu/videos
/dev/mapper/homework	ext4	755M	220K	700M	1%	/home/balu/homework
tmpfs	tmpfs	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	tmpfs	197M	4.0K	197M	1%	/run/user/1000

```
[root@archlinux balu]#
```

Figure 6: Screenshot of lsblk; df -hT

5. 快照真的好難喔

Reference:

- [LVM 系統快照設定教學 \(iThome\)](#)
- [Linux 解压缩 tar.zst 类型文件 \(CSDN\)](#)
- [How to keep or drop LVM snapshot \(Server Fault\)](#)

我以 root 進入 VM，並進行以下操作：

1. 將 /dev/vdc1 設為 pv，並且加到 nasahw2-main vg 中

```
1 pvcreate /dev/vdc1
2 vgextend nasahw2-main /dev/vdc1
```

2. 建立 snapshot 的 lv

```
1 lvcreate -L 1GiB -s -n backup /dev/nasahw2-main/course
```

3. 掛載 /dev/nasahw2-main/backup 到 /mnt/backup

```
1 mkdir -p /mnt/backup
2 mount /dev/nasahw2-main/backup /mnt/backup
```

4. lsblk 結果如 Figure 7 所示。

5. 進行備份到 tar:

```
1 tar -I zstd -cvf /home/balu/backup.tar.zst /mnt/backup
```

6. 卸載 /mnt/backup，並刪除 snapshot。

```
1 umount /mnt/backup
2 lvremove /dev/nasahw2-main/backup
```

7. 執行 lsblk; df -hT，結果如 Figure 8 所示。

```

[root@archlinux balu]# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
fd0                                2:0      1    4K  0 disk
sda                                8:0      0   32G  0 disk
├─sda1                             8:1      0   200M  0 part  /boot
└─sda2                             8:2      0  31.8G  0 part  /
sr0                                11:0     1  1024M  0 rom
zram0                             253:0     0    983M  0 disk  [SWAP]
vda                                254:0     0     1G  0 disk
├─vda1                             254:1     0  1022M  0 part
│   └─nasahw2--main-course-real    252:4     0     1G  0 lvm
│       ├──nasahw2--main-course    252:0     0     1G  0 lvm  /home/balu/course
│       └─nasahw2--main-backup    252:6     0     1G  0 lvm  /mnt/backup
└─vdb                               254:16    0     1G  0 disk
    ├─vdb1                         254:17    0  1022M  0 part
    │   └─nasahw2--main-homework    252:1     0   800M  0 lvm
    │       └─homework              252:3     0   784M  0 crypt /home/balu/homework
    └─nasahw2--main-course-real    252:4     0     1G  0 lvm
        ├──nasahw2--main-course    252:0     0     1G  0 lvm  /home/balu/course
        └─nasahw2--main-backup    252:6     0     1G  0 lvm  /mnt/backup
vdc                                254:32    0     2G  0 disk
├─vdc1                             254:33    0     2G  0 part
│   └─nasahw2--main-backup-cow    252:5     0     1G  0 lvm
│       └─nasahw2--main-backup    252:6     0     1G  0 lvm  /mnt/backup
vdd                                254:48    0    16G  0 disk
└─vdd1                             254:49    0    16G  0 part
vde                                254:64    0    512M  0 disk
├─vde1                             254:65    0    510M  0 part
│   └─nasahw2--secondary-videos    252:2     0    508M  0 lvm  /home/balu/videos
vdf                                254:80    0     7G  0 disk
├─vdf1                             254:81    0     2G  0 part
├─vdf2                             254:82    0     2G  0 part
└─vdf3                             254:83    0     2G  0 part
vdg                                254:96    0     7G  0 disk
├─vdg1                             254:97    0     2G  0 part
├─vdg2                             254:98    0     2G  0 part
└─vdg3                             254:99    0     2G  0 part
vdh                                254:112   0     7G  0 disk
├─vdh1                             254:113   0     2G  0 part
├─vdh2                             254:114   0     2G  0 part
└─vdh3                             254:115   0     2G  0 part
vdi                                254:128   0     6G  0 disk
├─vdi1                             254:129   0     2G  0 part
└─vdi2                             254:130   0     4G  0 part  /mnt/myusb
[root@archlinux balu]#

```

Figure 7: Screenshot of lsblk after creating snapshot

```

[root@archlinux balu]# lsblk; df -hT
NAME                                MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
fd0                                2:0      1   4K  0 disk
sda                                8:0      0  32G  0 disk
├─sda1                             8:1      0 200M  0 part  /boot
├─sda2                             8:2      0 31.8G  0 part  /
sr0                               11:0      1 1024M  0 rom
zram0                             253:0      0 983M  0 disk  [SWAP]
vda                               254:0      0   1G  0 disk
├─vda1                             254:1      0 1022M  0 part
│ └─nasahw2--main-course           252:0      0   1G  0 lvm    /home/balu/course
vdb                               254:16     0   1G  0 disk
├─vdb1                             254:17     0 1022M  0 part
│ └─nasahw2--main-course           252:0      0   1G  0 lvm    /home/balu/course
│   └─nasahw2--main-homework       252:1      0 800M  0 lvm
│     └─homework                   252:3      0 784M  0 crypt /home/balu/homework
vdc                               254:32     0   2G  0 disk
├─vdc1                             254:33     0   2G  0 part
vdd                               254:48     0  16G  0 disk
├─vdd1                             254:49     0  16G  0 part
vde                               254:64     0  512M  0 disk
├─vde1                             254:65     0  510M  0 part
│ └─nasahw2--secondary-videos     252:2      0 508M  0 lvm    /home/balu/videos
vdf                               254:80     0   7G  0 disk
├─vdf1                             254:81     0   2G  0 part
├─vdf2                             254:82     0   2G  0 part
├─vdf3                             254:83     0   2G  0 part
vdg                               254:96     0   7G  0 disk
├─vdg1                             254:97     0   2G  0 part
├─vdg2                             254:98     0   2G  0 part
├─vdg3                             254:99     0   2G  0 part
vdh                               254:112    0   7G  0 disk
├─vdh1                             254:113    0   2G  0 part
├─vdh2                             254:114    0   2G  0 part
├─vdh3                             254:115    0   2G  0 part
vdi                               254:128    0   6G  0 disk
├─vdi1                             254:129    0   2G  0 part
├─vdi2                             254:130    0   4G  0 part  /mnt/myusb
Filesystem                        Type      Size  Used Avail Use% Mounted on
dev                              devtmpfs  976M    0  976M   0% /dev
run                              tmpfs     984M  740K  983M   1% /run
/dev/sda2                       ext4      32G   6.6G   23G  23% /
tmpfs                           tmpfs     984M    0  984M   0% /dev/shm
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                           tmpfs     984M    0  984M   0% /tmp
/dev/vdi2                       ntfs3     4.0G   22M   4.0G   1% /mnt/myusb
/dev/sda1                       vfat      197M   69M  129M  35% /boot
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-cryptsetup@homework.service
/dev/mapper/nasahw2--main-course ext4      459M   4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4      466M   66M  371M  16% /home/balu/videos
/dev/mapper/homework            ext4      755M  220K  700M   1% /home/balu/homework
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs                           tmpfs     197M   4.0K  197M   1% /run/user/1000

```

Figure 8: Screenshot of `lsblk; df -hT` after removing snapshot

6. 好老舊喔

Reference:

- [Removing Physical Volumes from a Volume Group \(Red Hat\)](#)
- [pvmove 命令 \(LinuxCool\)](#)
- [Remove a PV from LVM \(Stack Exchange\)](#)

我做的步驟如下：

1. 將 /dev/vdd1 設為 pv，並且加到 nasahw2-secondary vg 中

```
1 pvcreate /dev/vdd1
2 vgextend nasahw2-secondary /dev/vdd1
```

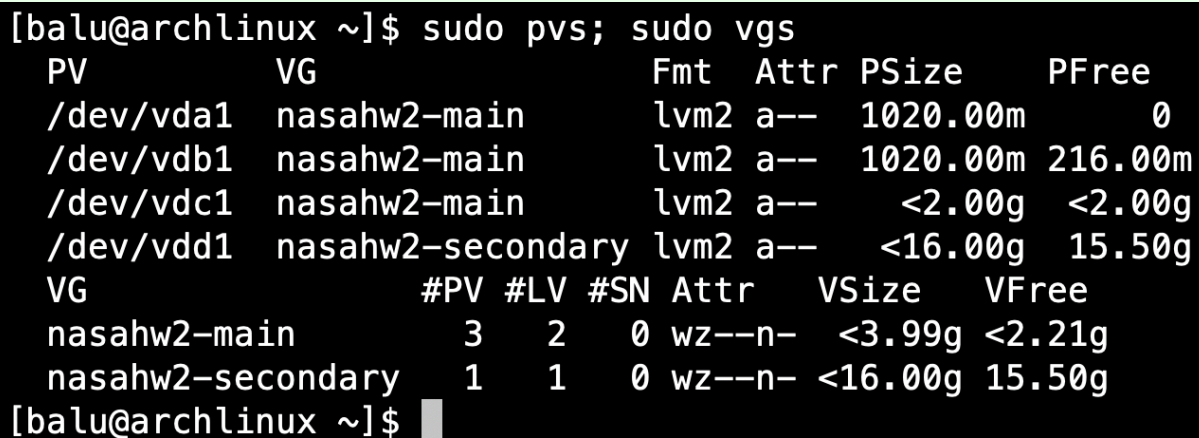
2. 搬移 pv /dev/vde1 中的內容到 /dev/vdd1

```
1 pvmove /dev/vde1 /dev/vdd1
```

3. 將 /dev/vde1 從 nasahw2-secondary vg 中移除，並且刪除 pv

```
1 vgreduce nasahw2-secondary /dev/vde1
2 pvremove /dev/vde1
```

Ans.



```
[balu@archlinux ~]$ sudo pvs; sudo vgs
PV          VG          Fmt Attr PSize  PFree
/dev/vda1   nasahw2-main  lvm2 a-- 1020.00m  0
/dev/vdb1   nasahw2-main  lvm2 a-- 1020.00m 216.00m
/dev/vdc1   nasahw2-main  lvm2 a--  <2.00g  <2.00g
/dev/vdd1   nasahw2-secondary lvm2 a--  <16.00g 15.50g
VG          #PV #LV #SN Attr   VSize  VFree
nasahw2-main      3  2  0 wz--n- <3.99g <2.21g
nasahw2-secondary 1  1  0 wz--n- <16.00g 15.50g
[balu@archlinux ~]$
```

Figure 9: Screenshot of `sudo pvs; sudo vgs`



7. 我看還是再來合一次吧

8. 等一下，妳還沒回答我